

Practice problems #12

1. For matrices A, B , and C below, show that $AB = AC$. What do you observe from this?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -8 \\ 2 & 3 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 5 & 2 \\ 1 & -2 \end{bmatrix}$$

2. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}$, and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$.

- (a) Is D an elementary matrix? Why or why not? Can you decompose D as $D = E_3 E_2 E_1$ where each E_i is an elementary matrix? If so, find them.
- (b) Compute AD and DA . Explain how the columns or rows of A change when A is multiplied by D on the right or on the left. Find a 3×3 diagonal matrix B (with $B \neq I$) such that $AB = BA$.
(A "diagonal matrix" is an $n \times n$ matrix with (i, j) -entry zero if $i \neq j$.)

3. Prove: If the columns of B are linearly dependent, then so are the columns of AB .

Answers are on page 2.

#1: $AB = AC = \begin{bmatrix} 7 & -2 \\ 21 & -6 \end{bmatrix}$. The case where $AB = AC$, yet $B \neq C$, is possible for matrices.

#2(a): D itself is not an elementary matrix. Why? Each E_i represents scaling. Find them.

#2(b): AD is obtained by multiplying the columns of A by 2,3,and 4, repsectively. DA is obtained by multiplying the rows of A by 2,3, and 4, respectively. Any diagonal matrix B such that $B = cI$ (where c is a scalar) will work.

#3: Hint: $Bx = 0$ has what kind of solutions?
 $(AB)x = 0$ has what kind of solutions?